

Ch1 Intro

- $P(A) = \#(A) / \#(\Omega)$
- $A \cap B = AB$ intersection (AND)
- $A \cup B$ union (OR)
- $AB = \emptyset$ disjoint, mutually exclusive, no overlap, no intersection
- ACB , inclusion, A occurs then so does B

Rules

- Addition: $P(A) = \sum_{i=1}^n P(A_i)$ if $A_1, \dots, A_n$ partition of A
- $P(\Omega) = 1$
- Complement: $P(A^c) = 1 - P(A)$
- Difference: $P(BA^c) = P(B) - P(A)$ if ACB
- Incl-Excl: $P(A \text{ or } B) = P(A) + P(B) - P(AB)$

- Relative Freq: # of times it happens / # of trials
 - long run: average = $P_n(A) = P(A)$ for large n
- Prob Dist over Ω
 - outcome $a, b, c, \dots$
 - prob $\{P(a), P(b), P(c), \dots\} \leftarrow$ sum to 1

Conditional Prob

- $P(A|B)$ = prob A given B, outcome space B
- chance of A if B is known to have occur
- $P(A^c|B) = 1 - P(A|B)$
- Rules
 - Division: $P(A|B) = \frac{P(AB)}{P(B)} = \frac{\#(AB)}{\#(B)}$ (counting)
 - Product: $P(AB) = P(A)P(B|A) = P(B)P(A|B)$
 - average: $P(A) = P(A|B_1)P(B_1) + \dots + P(A|B_n)P(B_n)$
 - Bayes: $P(B_i|A) = \frac{P(A|B_i)P(B_i)}{P(A)}$ (weighted avg)

Independence

- learning result of one trial doesn't affect chance
- e.g. 2 draw w/ replacement from box of known
- dependent: does affect chance for other
- e.g. 2 draw w/o replace, 2 draw w/ replace from box of card

$P(AB) = P(A)P(B) \iff P(A|B) = P(A) \iff P(B|A) = P(B)$ (doesn't affect chances)

And/or

- $P(A \cup B) = P(A) + P(B) - P(AB)$
- $P(AB) = P(A)P(B) - P(A \cap B)$
- $P(A \cap B) = P(A)P(B|A) = P(B)P(A|B)$
- $P(A \cap B^c) = P(A) - P(AB)$
- $P(B \cap A^c) = P(B) - P(AB)$

IND 2

- $P(AB) = P(A)P(B)$
- $P(A \cup B) = P(A) + P(B)$
- ME: $P(AB) = \{ \emptyset \}$ not both @ a time e.g. rolling 1 or 2

Ch2: Repeated trials, sampling Bernoulli (1, p)

Binomial

- $P(k \text{ success in } n \text{ trials}) = \binom{n}{k} p^k q^{n-k}$ ind trials
- Fixed n, k from 0 to n $\rightarrow \text{Bin}(n, p)$
- Probs add up to: $(p+q)^n = \sum_{k=0}^n \binom{n}{k} p^k q^{n-k}$
- $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ = n choose k
 - # ways to pick k places out of n
 - # subsets of k in set of n
 - # in row n, column k of Pascal's
 - $\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}$
- consecutive ratio
 - Pascal's: $\binom{n}{k} / \binom{n}{k-1} = \frac{n-k+1}{k}$
- $\text{Bin}(n, p) = \frac{P(k) - P(k-1)}{P(k-1) - P(k-2)} = \frac{(n-k+1)p}{kq}$

- mode $\text{Bin}(n, p) = m$ = most likely value
 - int $(np \pm p) \in [(n+1)p]$
 - $(n+1)p = \text{int} = 2 \text{ modes } m, m-1$
 - $(n+1)p = \text{not int} = m$
- normal approx. to Bin. Dist.
 - $P(a \leq b) \approx \Phi\left(\frac{b + \frac{1}{2} - \mu}{\sigma}\right) - \Phi\left(\frac{a - \frac{1}{2} - \mu}{\sigma}\right)$
 - only if $\sigma \geq 3$; $0 < \mu \pm 3\sigma < n$
 - $\mu = np$; $\sigma = \sqrt{npq}$

- Poisson approx
 - Note: large p, μ opposite ex: $P(\leq 76 \text{ fridge})$ of $\text{Bin}(100, .97)$ = $P(\leq 2 \text{ fridge})$ of $\text{Bin}(100, .03)$ = $P(0) + P(1) + P(2)$ w/ pois (μ)
 - small p, big n
 - $P(k) \approx \frac{e^{-\mu} \mu^k}{k!}$
- sampling w/ replacement
 - pop size N w/ g good & b bad & $N = G+B$
 - samp size $n = g+b$
 - $P(g \text{ good \& } b \text{ bad}) = \left(\frac{g}{N}\right)^g \left(\frac{b}{N}\right)^b = \left(\frac{g}{N}\right)^g \left(\frac{b}{N}\right)^b = X \sim \text{Bin}(n, \frac{g}{N})$

- sampling w/o replacement
 - $P(g \text{ good \& } b \text{ bad}) = \frac{\binom{g}{g} \binom{b}{b}}{\binom{N}{n}} = X \sim \text{Hyp}(n, N, G)$

- HV @ 13 card drawn
 - a) no court = $\frac{\binom{10}{13} \binom{4}{0}}{\binom{52}{13}}$
 - b) ≥ 1 ace, no other court cards = $\frac{\binom{4}{1} \binom{10}{12}}{\binom{52}{13}}$

- a) at most one kind of a court card
 - $\frac{\binom{4}{1} \binom{10}{12}}{\binom{52}{13}} + \frac{\binom{4}{2} \binom{10}{11}}{\binom{52}{13}} + \frac{\binom{4}{3} \binom{10}{10}}{\binom{52}{13}}$
- b) straight flush 1 suit
 - $\frac{\binom{4}{1} \binom{10}{12}}{\binom{52}{13}}$

- Sec 3
 - Given 12 heads in 20 tosses
 - first 2 lands heads = $\frac{\binom{2}{2} \binom{18}{10}}{\binom{20}{12}} = \frac{1}{10}$
 - 8 bets before stopping (win 5 to stop)
 - $\frac{\binom{8}{8} \binom{12}{0}}{\binom{20}{12}} = \frac{1}{10}$

Quiz 1

- IND, IN-EX: $P(\text{flu}) = 50\%$ & $P(\text{either flu or covid}) = 70\%$
- a) If getting flu & getting covid are IND, what is $P(\text{covid})$?
 - $P(A \cap B) = P(A)P(B)$
 - $P(\text{flu} \cap \text{covid}) = P(\text{flu})P(\text{covid}) \leftarrow \text{IND}$
 - $P(A \text{ or } B) = P(A) + P(B) - P(AB)$
 - $0.7 = 0.5 + P(B) - 0.5P(B)$
 - $P(B) = 0.4$
- b) $P(\text{either flu, covid, pox}) = 80\%$. Find $P(\text{pox})$ if mutually IND.
 - $P(A \cup B \cup C) = 0.5 + 0.4 + C - 0.2 - 0.5C - 0.4C + (0.4 \times 0.5 \times C)$
 - $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(AB) - P(AC) - P(BC) + P(ABC)$

- 2 Cond, MULT, ADD: Box w/ n balls (2, 3, 4) & 1 red, n-1 blue
 - a) $P(\text{blue} | n=3) = \frac{2}{3}$ (forward)
 - b) $P(n=3 | \text{blue}) = \frac{P(3 \text{ balls \& blue})}{P(\text{blue})} = \frac{(\frac{1}{3})(\frac{2}{3})}{(\frac{1}{3})(\frac{2}{3}) + (\frac{2}{3})(\frac{1}{3})} = \frac{1}{2}$
 - c) $P(n = \text{most likely} | \text{blue}) = \text{calculate for all 3} \leftarrow \frac{P(A|B) + P(B|A)P(A)}{P(B)}$

- 3 Bin, HG: cards & die
 - a) Draw 8 card poker: 2 three of a kinds, no rank repeat (KKKQQQ47) = $\frac{\binom{4}{3} \binom{4}{3} \binom{4}{2} \binom{4}{1}}{\binom{52}{8}}$
 - b) Roll 20 times: 0 5 in first 9, 6 in 10th, 3 5's & no 6 in next 10
 - $P(\text{no 5s in } 9\text{'s}) \times P(6) \times P(3 \text{ 5's, no 6 in next } 10) = \left(\frac{9}{10}\right)^9 \times \frac{1}{10} \times \left(\frac{9}{10}\right)^3 \left(\frac{1}{10}\right)^7$
 - $\frac{9^9}{10^{10}} \times \frac{1}{10} \times \frac{9^3}{10^7} = \frac{9^{12}}{10^{18}}$
 - c) Roll 20 times: given 1st 6 on 10th roll, $P(5 \geq 3)$
 - $P(5 \text{ in first } 9) + P(5 \text{ on } 10\text{th}) + P(5 \text{ in last } 10)$
 - $\frac{\binom{9}{5} \binom{10}{5}}{\binom{20}{10}} + \frac{\binom{10}{5} \binom{10}{5}}{\binom{20}{10}}$

- 4 APPR: 2500 customers. Normal day, ea. customer has $P(\text{call company}) = 0.002$. X = # calls recieved on a day
 - a) Est. $P(\geq 1 \text{ customer calls on a day})$
 - large $n = 2500$ $\Rightarrow P(X \geq 1) = 1 - P(X=0)$
 - small $p = 0.002 \Rightarrow 1 - 5^{-5} = 1 - e^{-5}$
 - $\mu = np = 5$
 - check $\mu \geq 30$
 - b) Use pois to est. p of customer calling for chance $P(X=0) \geq 0.95$
 - $P(X=0) = \frac{\mu^0 e^{-\mu}}{0!} = e^{-\mu} \geq 0.95 \Rightarrow \mu \leq -\log(0.95)$
 - $\mu = 500$ { $\mu \pm 3\sigma < n$ valid approx $\Rightarrow 1 - \Phi\left(\frac{515 - 5 - 500}{20}\right)$
 - c) $p = 0.2$. $P(\text{no more than } 925 \text{ calls})$
 - $\mu = 500$ { $\mu \pm 3\sigma < n$ valid approx $\Rightarrow 1 - \Phi\left(\frac{925 - 500 - 500}{20}\right)$

Form A.1

- 1 coin 1, 2, 3
 - a) $P(HH | \text{coin } 3) = \frac{1}{4} \times \frac{1}{4}$ (forward)
 - b) $P(HH) = P(HH \& (1)) + P(HH \& (2)) + P(HH \& (3)) = \frac{1}{2}(\frac{1}{2} \times \frac{1}{2}) + \frac{1}{3}(\frac{1}{2} \times \frac{1}{2}) + \frac{1}{4}(\frac{1}{2} \times \frac{1}{2})$
 - c) $P(\text{coin } 2 | HH) = \frac{P(HH | \text{coin } 2)P(\text{coin } 2)}{P(HH)} = \frac{\frac{1}{2} \times \frac{1}{3}}{\frac{1}{2} \times \frac{1}{3} + \frac{1}{4} \times \frac{1}{4} + \frac{1}{2} \times \frac{1}{2}}$
 - d) If HH, which coin most likely to have picked?
 - find all 3 $\Rightarrow i=1$ gives largest probability

- 2 Roll die 15 times
 - a) $P(\geq \text{one } 3, \geq \text{one } 4, \geq \text{one } 5, \text{ or } \geq \text{one } 6) = 1 - P(1 \text{ or } 2) = 1 - (\frac{1}{6} + \frac{1}{6})^{15}$
 - b) $P(\text{no } 3\text{'s, no } 4\text{'s, no } 5\text{'s, or no } 6\text{'s}) = P(A \cup B \cup C \cup D) = P(A) + P(B) + P(C) + P(D) - (P(AB) + P(AC) + P(AD) + P(BC) + P(BD) + P(CD)) + P(ABCD)$
 - c) $P(\text{no } 3\text{'s, no } 4\text{'s, no } 5\text{'s, or no } 6\text{'s}) = 1 - P(\text{at least one } 3, 4, 5, 6) = 1 - (1 - (\frac{1}{6})^{15}) = (\frac{1}{6})^{15}$

- 2/11 ADJ: Draw 5 cards w/o replacement. Given you did NOT get exactly 2 queens, find p you got ≥ 1 King
 - $P(A) = P(\geq 1 \text{ King}) \rightarrow P(A^c) = P(\text{no King})$
 - $P(B) = P(\text{exactly 2 queens}) \rightarrow P(B^c) = P(\text{no queen})$
 - $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) + P(B) - P(A \cup B)}{P(B)} = \frac{1 - P(A^c) - P(B^c) - P(A^c \cap B^c)}{P(B)}$
 - $P(A|B) = \frac{1 - \frac{\binom{4}{0} \binom{48}{5}}{\binom{52}{5}} - \frac{\binom{4}{2} \binom{48}{3}}{\binom{52}{5}} - \frac{\binom{4}{3} \binom{48}{2}}{\binom{52}{5}}}{\frac{\binom{4}{2} \binom{48}{3}}{\binom{52}{5}}}$

- 2/4 APJ: Consec. Odds
 - $P(K) = \frac{P(K)}{P(K-1)} \cdot P(1-p)^{K-1} \cdot p \rightarrow \text{max when sign change}$
 - a) $P(K) = \frac{P(K)}{P(K-1)} \cdot \dots = \frac{K-1}{K-4} (1-p)$... $P(7) < P(8) < P(9) \dots$
 - b) $\frac{K-1}{K-4} (1-p) > 1 \Rightarrow K > \frac{p+3}{p}$ for $R(K) > 1 \Rightarrow P(K) > P(K-1)$
 - c) $P(K) = \frac{e^{-\mu} \mu^K}{K!} \Rightarrow P(K) = \frac{\mu}{K}$ & double max when K is integer

- G1 Review
 - 23 TS songs & 15 SC songs. Pick 12 songs, $P(8-9 \text{ TS songs}) = \frac{\binom{23}{8} \binom{15}{4}}{\binom{38}{12}} + \frac{\binom{23}{9} \binom{15}{3}}{\binom{38}{12}}$
 - lec: 5 card poker hand
 - $P(4 \text{ aces \& } 1 \text{ king}) = \frac{\binom{4}{4} \binom{1}{1} \binom{46}{3}}{\binom{52}{5}}$
 - $P(4 \text{ aces}) = \frac{\binom{4}{4} \binom{48}{1}}{\binom{52}{5}}$
 - $P(4 \text{ of a kind}) = \frac{\binom{13}{1} \binom{4}{4} \binom{48}{1}}{\binom{52}{5}}$
 - $P(2 \text{ of a kind}) = \frac{\binom{13}{2} \binom{4}{2} \binom{4}{1} \binom{4}{1}}{\binom{52}{5}}$
 - $P(2 \text{ of a kind, 2 jingles}) = \frac{\binom{13}{2} \binom{4}{2} \binom{4}{1} \binom{4}{1}}{\binom{52}{5}}$

Section 2.3 Q4 Review

Q1: $N \sim \text{pois}(\mu)$

- $\mu_1, \mu_2, \dots$ i.i.d. unif(0,1) ind of N
- $\text{MIN} = \min(\mu_1, \dots, \mu_n)$

① If $\mu_1, \dots, \mu_n$ i.i.d. $U(0,1) \Rightarrow U(x) \sim \text{Beta}(k, n+1-k)$

$\text{MIN} \sim \text{Beta}(1, n+1-1) \Rightarrow E(\text{Beta}(r,s)) = \frac{r}{r+s}$

$E(\text{MIN}) = \frac{1}{1+n}$

② $E(N) = E(E(\text{MIN}))$

$= E\left(\frac{1}{1+n}\right) = \sum_{n=0}^{\infty} \frac{1}{1+n} \cdot \frac{e^{-\mu} \mu^n}{n!}$

Q2: X, Y i.i.d. RV w/ $f_X(x) = \frac{1}{x^2}, x > 1$. Find $Z = \frac{Y}{X}$

Convolution: $(X, Y) \rightarrow (X, Z)$

- $f_{X,Z}(x, z) = f_{X,Y}(x, y(x, z)) \left| \frac{\partial y}{\partial z} \right|$
- $f_Z(z) = \int_{-\infty}^{\infty} f_{X,Z}(x, z) dx$

① $f_{X,Y}(x, y) = f_X(x) f_Y(y) = \frac{1}{x^2 y^2}, x > 1, y > 1$

② $Z \rightarrow$ find $Y \rightarrow$ derive: $Z = \frac{Y}{X} \rightarrow Y = ZX \rightarrow \frac{\partial y}{\partial z} = X$

③ $f_{X,Y}(x, z) = \frac{1}{x^2 (zx)^2} X = \frac{1}{x^3 z^2}, x \geq 1, xz \geq 1 \Rightarrow x \geq \frac{1}{z}$

④ Bounds + int:

- $x \geq \frac{1}{z}$
- $x > 1, y > 1 \Rightarrow Z$ pos $\rightarrow (Z > 0)$
- $(Z \geq 1) \Rightarrow \frac{1}{z} \geq 1 \Rightarrow z \leq 1$

$f_Z(z) = \int_{\frac{1}{z}}^{\infty} \frac{1}{x^3 z^2} dx = \frac{1}{z^2} \left[-\frac{1}{2x^2} \right]_{\frac{1}{z}}^{\infty} = \frac{1}{2z^2}$

$\Rightarrow f_Z(z) = \begin{cases} \frac{1}{2z^2}, & 0 < z < 1 \\ 0, & \text{else} \end{cases}$

Q3: $Z_1, \dots, Z_4 \sim N(0,1), V = \sqrt{Z_1^2 + \dots + Z_4^2}$

$Z \sim N(0,1) \Rightarrow Z^2 \sim \chi^2_1 \sim \text{Gamma}(\frac{1}{2}, \frac{1}{2})$

$X \sim \text{Gamma}(\alpha, \lambda), Y \sim \text{Gamma}(\beta, \lambda)$ ind.

$\Rightarrow X+Y \sim \text{Gamma}(\alpha+\beta, \lambda)$

$X \sim \text{Gamma}(\alpha, \lambda)$

$f_X(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, x > 0$

① $Z_1 \sim N(0,1) \Rightarrow Z_1^2 \sim \chi^2_1 \sim \text{Gamma}(\frac{1}{2}, \frac{1}{2})$

$V^2 = Z_1^2 + \dots + Z_4^2 \sim \text{Gamma}(\frac{1}{2} \cdot 4, \frac{1}{2})$

$\Rightarrow V^2 \sim \text{Gamma}(2, \frac{1}{2})$

② Find density of V

- A red balls, B blue balls
- Pick half of balls at random
- X1A = # red out of A picked up

① Pick w/o replacement:

$X1A \sim \text{HG}(n-A, N=2A, q=A)$

② Pick w/o replacement: A-poisson

$E(X) = E(E(X1A)) = E\left(\frac{A}{2}\right) = \frac{A}{2}$

③ w/ replacement: $X1A \sim \text{Bin}(n=A, \frac{1}{2})$

$\text{Var}(X) = \text{Var}(E(X1A)) + E(\text{Var}(X1A))$

$= \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$

FAG

Q2: $X \sim N(2,4), Y \sim N(1,3)$, ind. $P(2X \geq Y+2) = ?$

$2X - Y \geq 2$

$Z = 2X - Y \sim N(2(2)-1, 2^2(4)+3) = N(3,19)$

$P(Z \geq 2) = 1 - P(Z < 2)$

$= 1 - \Phi\left(\frac{2-3}{\sqrt{19}}\right)$

Q1: $f_{X,Y}(x,y) = \begin{cases} \frac{3}{8}x, & \text{if } x+y \geq 2, x < 2, y < 2 \\ 0, & \text{else} \end{cases}$

not ind.

Y-limits: $2 \leq y < x-1$

1.1 $P(X > Y+1)$

$= \int_{1/2}^2 \int_{2-x}^x \frac{3}{8} x dy dx$

1.2 Find marginal of Y

$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx = \int_{2-y}^2 \frac{3}{8} x dx$

1.3 not ind. b/c $"x+y \geq 2"$

1.4 U, V jointly cont. w/ $U+V$ cont. w/ pdf:

$f_{U+V}(z) = \int_{-\infty}^{\infty} f_{U,V}(u, z-u) du \Leftarrow f_{U,V}(z)$

$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_{X,Y}(x, z-x) dx$

$= \int_{1/2}^2 \frac{3}{8} x dx$

Prac Q4

Q1: $X \sim \text{Gamma}(\alpha, \gamma), Y \sim \text{Gamma}(\beta, \gamma)$ ind.

a) Show $f_Z(z) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \frac{z^{\alpha+\beta-1}}{(1+z)^{\alpha+\beta}} \Leftarrow$ Convolution!

$f_{X,Y}(x,y) = \frac{\gamma^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\gamma x} \cdot \frac{\gamma^\beta}{\Gamma(\beta)} y^{\beta-1} e^{-\gamma y}$

$f_Z(z) = \int_0^z \frac{\gamma^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\gamma x} \cdot \frac{\gamma^\beta}{\Gamma(\beta)} (z-x)^{\beta-1} e^{-\gamma(z-x)} dx$

$= \frac{\gamma^{\alpha+\beta}}{\Gamma(\alpha)\Gamma(\beta)} e^{-\gamma z} \int_0^z x^{\alpha-1} (z-x)^{\beta-1} dx$

$= \frac{\gamma^{\alpha+\beta}}{\Gamma(\alpha)\Gamma(\beta)} \cdot \frac{z^{\alpha+\beta-1}}{(1+z)^{\alpha+\beta}} \cdot \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$

Integrals:

- Gamma: $\int_0^{\infty} x^{k-1} e^{-\lambda x} dx = \frac{\Gamma(k)}{\lambda^k}$
- Beta: $\int_0^1 t^{a-1} (1-t)^{b-1} dt = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$
- Normal: $\int_{-\infty}^{\infty} e^{-kx^2} dx = \sqrt{\frac{\pi}{k}}$

Q2: Throw 10 darts on x,y plane $\xi, \eta, \gamma \sim N(0,1)$

Find Prob that each dart is within 3.45 units from (0,0)

- Distance of each dart from origin: $D_i \sim \text{Rayleigh}$
- Each dart ind: raise to 10th power
- $P(\text{all 10 } 3-5 \text{ units away}) = P(3 \leq D_i \leq 5)^{10}$
- $= (F_{D_i}(5) - F_{D_i}(3))^{10}$
- $= (e^{-(\frac{1}{2} \cdot 3^2)} - e^{-(\frac{1}{2} \cdot 5^2)})^{10}$

Q3: $X, Y \sim N(0,6^2)$ ind.

a) $X+Y \sim N(0, 2 \cdot 6^2)$

b) Prove $(X+Y)^2 \sim \text{Gamma}$

To get standard normal: Subtract mean, divide by SD

$\frac{X+Y-0}{\sqrt{2 \cdot 6^2}} = \frac{X+Y}{2 \cdot 6}$

$\frac{(X+Y)^2}{2 \cdot 6^2} \sim \chi^2_1 = \text{Gamma}(\frac{1}{2}, \frac{1}{2 \cdot 6^2})$

$Z = X+Y \sim N(0, 4 \cdot 6^2)$

$f_Z(z) = \frac{1}{\sqrt{4 \cdot 6^2}} e^{-\frac{z^2}{4 \cdot 6^2}}$

$f_W(w) = \sum_{z \in W} f_Z(z) \left| \frac{dz}{dw} \right| = f_Z(\sqrt{w}) \frac{1}{2\sqrt{w}} + f_Z(-\sqrt{w}) \frac{1}{2\sqrt{w}}$

$= \frac{2}{2\sqrt{w}} \cdot \frac{1}{\sqrt{4 \cdot 6^2}} e^{-\frac{w}{4 \cdot 6^2}} = \frac{1}{\sqrt{w}} \cdot \frac{1}{\sqrt{4 \cdot 6^2}} e^{-\frac{w}{4 \cdot 6^2}}$

$\Rightarrow \text{gamma}(\frac{1}{2}, \frac{1}{4 \cdot 6^2})$

Rev session

$\text{dist } X_{(1)} - X_{(2)} \sim \text{Beta}(2,3) \Rightarrow E(X_{(1)} - X_{(2)}) = \frac{2}{5}$

$E(Y) = E(E(Y|X))$

$\text{Var}(Y) = E(\text{Var}(Y|X)) + \text{Var}(E(Y|X))$

$\text{Var}(E(Y|X)) = E(E(E(Y|X)^2)) - (E(E(Y|X)))^2 = E(E(Y^2|X)) - (E(Y))^2$

Rayleigh:

- If $Z \sim \text{Exp}(\frac{1}{2}) \rightarrow R = \sqrt{Z}, R \sim \text{Rayleigh}$
- $X, Y \sim N(0,1) \rightarrow X^2, Y^2 \sim \text{Gamma}(\frac{1}{2}, \frac{1}{2})$
- $X^2 + Y^2 \sim \text{Gamma}(1, \frac{1}{2}) = \text{Exp}(\frac{1}{2}) \rightarrow \sqrt{X^2 + Y^2} = R$

$f_Z(z) = \frac{1}{2} e^{-\frac{z^2}{2}} \rightarrow f_R(r) = \left| \frac{dr}{dz} \right| f_Z(z) = 2r e^{-\frac{1}{2} r^2}, r > 0$

$\Rightarrow F_R(r) = P(R < r) = P(R^2 < r^2)$

$= P(Z < r^2) = 1 - e^{-\frac{1}{2} r^2}, r > 0$

X, Y ind. $\sim N(\mu, \sigma^2)$

- $E(ax+by) = aE(x) + bE(y) = a\mu + b\mu$
- $\text{Var}(ax+by) = a^2 \text{Var}(x) + b^2 \text{Var}(y) = a^2 \sigma^2 + b^2 \sigma^2$

Convolution:

- Use change of variable for $f(x,y) \rightarrow f(x,z)$
- transforming joint dist
- Plug in y as a func of x and z for $f_{X,Y}(x,y)$

Find marginal of Z by integrating $f(x,z)$ w/ respect to y

- Set integral bounds as func of Z

$D = \sqrt{X^2 + Y^2}$

$X, Y \sim N(0,1) \rightarrow D = \sqrt{X^2 + Y^2} \sim \text{Rayleigh}$

$P(D \geq 5) = \int_5^{\infty} D e^{-D^2/2} dD = P(D^2 \geq 25) \rightarrow D^2 \sim \text{Exp}(\frac{1}{2}), t = D^2$

$\int_5^{\infty} \frac{1}{2} e^{-\frac{t}{2}} dt = \left[-e^{-\frac{t}{2}} \right]_{25}^{\infty} = e^{-\frac{25}{2}} = e^{-12.5}$

TR Q4 REV

Q1) shoot at target (0,0) $(X,Y) =$ where it hits; X, Y ind.

$X, Y \sim N(0, \sigma^2)$

a) $P(\text{arrow win 1 unit of center } \sigma=1)$

$P(\sqrt{X^2 + Y^2} \leq 1)$

Std. Rayleigh: $F_{\sqrt{X^2 + Y^2}}(1) = 1 - e^{-\frac{1^2}{2}}$

$P(X^2 + Y^2 \leq 1) = 1 - e^{-\frac{1}{2}}$

$\uparrow \text{Exp}(\frac{1}{2})$

b) $P(|X|+|Y| \leq 1)$

$\int_{\text{square}} \rightarrow \#$

rotation = joint variance

$\left[\Phi\left(\frac{1}{\sqrt{2}}\right) - \Phi\left(-\frac{1}{\sqrt{2}}\right) \right]^2$

Q3) $a < b < n$ positive: X, Y a th + b th order stat on n i.i.d. $\text{unif}(0,1)$

$\Rightarrow$ joint: $f_{X,Y}(x,y) = c(y-x)^{a-1} (c \leq x \leq y \leq 1), c > 0$

a) a, b, n, c

$X^{a-1} (Y-X)^{b-a-1} (1-Y)^{n-b-1} (0 \leq x \leq y \leq 1)$

$\Rightarrow a=1, b=7, n=7$

$\Rightarrow c = (a-1, 1, b-a-1, a, n-b) = (0, 1, 5, 1, 0) = 42$

b) range = marginal of Y

$f_Y(y) = \int f_{X,Y}(x,y) dx = \int_0^y 42(y-x)^5 dx$

OR: y is b th order stat

$Y \sim \text{Beta}(7, 1)$ on $0 < Y < 1$

Q4

Q1: $f_X(x) = \begin{cases} \frac{1}{x^2}, & x > 1 \\ 0, & \text{else} \end{cases}$. Find pdf Y

$Y = e^X$

change of var: $f_Y(y) = \frac{f_X(x)}{\left| \frac{dx}{dy} \right|} = f_X(x) \left| \frac{dx}{dy} \right|$

$Y = e^X \Rightarrow \log(Y) = X \Rightarrow \frac{dx}{dy} = \frac{1}{y}$

$f_Y(y) = \frac{1}{y^2} \cdot \frac{1}{y} = \frac{1}{y^3}, y > e$

$= \frac{1}{(\log(y))^3} \cdot \frac{1}{y}, y > e$

$0, \text{ else}$

Q2: $X \sim \text{unif}(1,5), Y \sim \text{unif}(X,5)$

$E(Y) = E(E(Y|X)) = E\left(\frac{X+5}{2}\right) = E\left(\frac{X}{2} + \frac{5}{2}\right) = \frac{E(X)}{2} + \frac{5}{2}$

unif mean $\frac{a+b}{2} = \frac{1+5}{2} = 3$

$= \frac{3}{2} + \frac{5}{2} = 4$

Q3: $U_1, \dots, U_5 \sim \text{unif}(0,1)$ w/ $X = U_1, Y = U_5$

Show: $f_{X,Y}(x,y) = 20x^3$ if $0 < x < y < 1$, 0 else.

$\frac{d^5}{dx^5} \frac{d^5}{dy^5} \int_0^1 \int_0^1 (1-x)^4 (1-y)^4 dx dy = 20x^3 dx dy$

X Btw 0,1 b/c X 2nd largest of 5 largest

$U \sim \text{unif}(0,1)$ PDF 1 4 outside bounds 0

Q4: U, V jointly cont. so $V-U$ cont. w/ pdf:

$f_{V-U}(z) = \int_{-\infty}^{\infty} f_{U,V}(u, u+z) du$

$f_{Y-X}(z) = \int_0^{1-z} 20x^3 dx = 5(1-z)^4, 0 \leq z < 1$

$0, \text{ else}$

$f_{X,Y}(x,y) = 20x^3, 0 < x < y < 1$

$0 < x < 1$

$0 < x+y < 1$

$0 < x < 1-z$

$Y \in (x,1)$

$X \in (0,Y)$

Discrete Distribution Summary (Pg 476)

Name	$p(k) = p\{X=k\}$	mean	var(X)
uniform: equally likely outcome	$\frac{1}{b-a+1}$	$\frac{a+b}{2}$	$\frac{(b-a+1)^2 - 1}{12}$

Bernoulli(p): 0,1

$P(1) = p$			
$P(0) = 1-p$	p		$p(1-p)$

Binomial (n, p) : # k success in n fixed trials

Poisson (μ): # arrivals in fixed time period/
approx Bin(n, p) if big n

hypergeometric (n, N, G) : # of
 G good in pop N w/o replacement
(got a good exam = success)

$$\frac{\binom{G}{k} \binom{N-G}{n-k}}{\binom{N}{n}} \quad \frac{nG}{N} \quad n \binom{G}{N} \left(\frac{N-G}{N} \right) \left(\frac{N-1}{N-1} \right)$$

geometric(p):

$\{1, 2, 3, \dots\}$ # of trials till 1 st success	$(1-p)^{x-1} p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
$\{0, 1, 2, \dots\}$ # of failures b4 1 st success	$(1-p)^x p$	$\frac{1-p}{p}$	$\frac{1-p}{p^2}$

negative binomial (r, p) :
 $\{0, 1, 2, \dots\}$ # of failures before r^{th} success $\binom{r+k-1}{r-1} p^r (1-p)^k \frac{r(1-p)}{p} \frac{r(1-p)}{p^2}$

Normal continuous: $\Phi\left(\frac{x - \mu}{\sigma}\right)$

$$\text{Var}(X) = E(X^2) - (E(X))^2 = \sum_{\text{all } x} x^2 P(X=x) - \left[\sum_{\text{all } x} x P(X=x) \right]^2$$

- whichever comes first: $X_B \sim \text{Geom}(p_1)$ $\left\{ \begin{array}{l} P(\min\{X_B, X_M\} > n) \\ P(X_B > n, X_M > n) = P(X_B > n) P(X_M > n) \\ = q_1^n q_2^n = (q_1 q_2)^n \end{array} \right. \quad q_i = 1 - p_i$
 $X_M \sim \text{Geom}(p_2)$

new $q = q_1, q_2 \rightarrow p = 1 - q_1, q_2$
 $\Rightarrow \text{Geom}(1 - q_1, q_2)$

Markov: $P(X > a) \leq \frac{E(X)}{a}$

Chebyshev: $IP(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$

- ① x is nonnegative
- ② $E(x)$ is known

- ① mean & var is known
- ② tail prob, tighter

Indicator Problems:

① Identify Indicator: $I_j = \begin{cases} 1 & A_j \text{ occurs} \\ 0 & \text{else} \end{cases}$ $\sum_{j=1}^n I_j = x \in \{0, 1, \dots, n\}$

"counting" problem: "number of"

③ choose indicator: I_j are distinct ways of success

e.g. $X = \# \text{ ways to get heads in 3 coin tosses}$ $I_{HHT}, I_{HTH}, I_{THH}$

③ Consider interaction: Ind? ME? Subject?

$$\text{Var}(X) \rightarrow E(X^2) \rightarrow E(I_1 I_1) = P(A_1 \cap A_1)$$

④ Formulas: $E(X) = \sum_{j=1}^n E(I_j) = \sum_{j=1}^n P(A_j)$

$$E(X^2) = \sum_{j=1}^n E(I_j) + \sum_{i \neq j} E(I_i I_j)$$

ME: 0

Subset: $\mathbb{P}\{A_j\}$

← pairs ≠ to each other

← n(n-1) terms

$$= \sum_{j=1}^n \mathbb{P}(A_j) + \sum_{i \neq j} \mathbb{P}(A_i \cap A_j)$$

Stop If: $\left. \begin{array}{l} \frac{n-1}{n-1} F \\ \frac{n-1}{n-1} S \end{array} \right\} \begin{array}{l} \binom{k-1}{n-1} p^{n-1} q^{k-n} p + \binom{k-1}{n-1} q^{n-1} p^{k-n} q \\ \binom{k-1}{n-1} p^n (1-p)^{k-n} + \binom{k-1}{n-1} (1-p)^n p^{k-n} \end{array}$

EX: n balls thrown ind. at random into b boxes. X = # of boxes left empty

$$I_1 = \begin{cases} 1 & \text{if box empty} \\ 0 & \text{else} \end{cases}$$

$$E(I_j) = P(j^{\text{th}} \text{ box empty}) = \underbrace{\left(\frac{b-1}{b}\right)^n}_{\text{indep}} \leftarrow \text{each throw}$$

$$X = \sum_{j=1}^b \Gamma_j$$

$$F(x) = \sum_{j=1}^b I_j = \sum_{j=1}^b \left(\frac{b-1}{b}\right)^j = b \left(\frac{b-1}{b}\right)^b$$

$$E(I, I_j) = E(I_{nj}) = P(\text{1st \& jth box empty}) = \left(\frac{b-2}{b}\right)^n$$

$$E(X^2) = \sum_{j=1}^b E(I_j) + \sum_{j=1}^b E(I_{in,j}) = b\left(\frac{b-1}{b}\right)^n + b(b-1)\left(\frac{b-2}{b}\right)^n$$

n pair chopsticks
✓ grab k pairs & try to insert
 $X = \#$ matching pairs

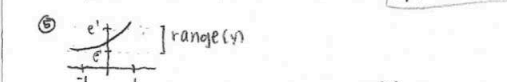
$$I_A = \begin{cases} 1 & \text{if a pair} \\ 0 & \text{not} \end{cases}$$

If $y = g(x)$ and g invertible: $f_Y(y) = \left| \frac{dg^{-1}(y)}{dy} \right| \cdot f_X(g^{-1}(y))$

- Define trans: $y = g(x)$
- Find inv: solve $x = g^{-1}(y)$ to go from y to x
- Find deriv inv: solve $\frac{dy}{dx} g^{-1}(y)$
- Apply Form: new PDF of y is $f_Y(y) = \dots$
- Range of y based on trans: $y = g(x)$

Ex: $X \sim \text{unif}(-1,1)$ and $Y = e^X$. Find density of y

- $f(x) = \frac{1}{b-a} = \frac{1}{1-(-1)} = \frac{1}{2}$, $x \in (-1,1)$, $y = e^x = g(x)$
- $g(x) = e^x$
 $g^{-1}(y) = \log y = \log e^y = y$
- $\frac{dy}{dx} g^{-1}(y) = \frac{dy}{dy} y = 1$
- $f_Y(y) = \frac{1}{2} \cdot \frac{1}{y} = \frac{1}{2y}$, $y \in (e^{-1}, e^1)$

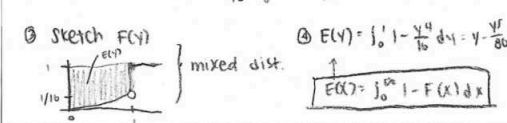


PDF $\rightarrow$ CDF verify: $f_Y(y) \geq 0$
 $\int f(x) \rightarrow F(x)$
 $f(x) \leftarrow \frac{d}{dx} F(x)$
 $\int_{e^{-1}}^e \frac{1}{2y} dy = \left[\frac{1}{2} \log y \right]_{e^{-1}}^e = \frac{1}{2} \log e^2 = 1$ ✓

Ex: 4 ppl meet at cafe in 2 hrs. Arrival $Z_1, \dots, Z_4 \stackrel{iid}{\sim} U(0,2)$.
 $X = \max(Z_1, \dots, Z_4)$, time last person arrives.

Find $f(x)$ density:
 $f(x) = \frac{1}{b-a} = \frac{1}{2}$, $x \in (0,2)$
 $P(X \in dx) = \left(\frac{4}{2} \right) \left(\frac{1}{2} \right)^3 \frac{1}{2} dx$
 $= 4 \left(\frac{x^3}{6} \right) dx = \frac{2}{3} x^3 dx$, $0 < x < 2$

Let $Y = \min(X, 1)$. Find $F_Y(y) \leftarrow$ CDF
 $F_Y(y) = P(Y \leq y) = \begin{cases} 0 & y \leq 0 \\ \frac{y^4}{16} & 0 < y < 1 \\ 1 & y \geq 1 \end{cases}$
 $P(Y \leq y) = P(X \leq y) = \int_0^y \frac{x^3}{4} dx = \frac{x^4}{16} \Big|_0^y = \frac{y^4}{16}$



Sketch $F_Y(y)$ } mixed dist.
 $E(Y) = \int_0^1 1 - \frac{y^4}{16} dy = 1 - \frac{y^5}{80} \Big|_0^1 = 1 - \frac{1}{80} = \frac{79}{80}$
 $E(X) = \int_0^{\infty} 1 - F(x) dx$

$S_X(t) = M_{\log(X)}(t)$; $X, Y \stackrel{iid}{\sim}$ w/ density $f(x) = \frac{1}{2x}$, $1 \leq x < \infty$
 Show $S_X(t) = \frac{1}{1-t}$, $t \neq 1$
 $S_X(t) = M_{\log(X)}(t) = E(e^{t \log(X)}) = E((e^{\log(X)})^t) = E(X^t)$
 $E(X^t) = \int_1^{\infty} x^{t-2} dx = \left[\frac{x^{t-1}}{t-1} \right]_1^{\infty} = -\frac{1}{t-1} = \frac{1}{1-t}$, $t \neq 1$

Find $S_{XY}(t)$
 Note: MGF: $S_{XY}(t) = M_{\log(XY)}(t) = E((XY)^t) = E(X^t Y^t) = E(X^t) E(Y^t) = \left(\frac{1}{1-t} \right)^2$
 Note: $M_{\text{Gamma}(r, \lambda)}(t) = \left(\frac{\lambda}{\lambda - t} \right)^r$
 $\log(X, Y) \sim \left(\frac{\lambda}{\lambda - t} \right)^2$

MGF: $M_X(t) = E(e^{tX}) = \int_{-\infty}^{\infty} e^{tx} f_X(x) dx$

$f_X(x) = \begin{cases} e^{-3x} + \frac{2}{3} e^{-x}, & x > 0 \\ 0, & \text{else} \end{cases}$
 a) $M_X(t) = E(e^{tX}) = \frac{1}{3-t} + \frac{2}{3(1-t)}$ for $t < 1$ show
 $H_X(t) = E(e^{tX}) = \int_0^{\infty} e^{tx} (e^{-3x} + \frac{2}{3} e^{-x}) dx$
 $= \frac{1}{t-3} e^{t \cdot \infty - 3 \cdot \infty} - \frac{2}{3(t-1)} e^{t \cdot \infty - 1 \cdot \infty} \Big|_{x=0}^{\infty}$
 $= 0 - \frac{1}{t-3} + \frac{2}{3(1-t)} = \frac{1}{3-t} + \frac{2}{3(1-t)}$
 $= \frac{1}{3-t} + \frac{2}{3(1-t)} \Rightarrow t \neq 1$

b) $E(X) = m'(0) = \frac{1}{(3-0)^2} + \frac{2}{3(1-0)^2} = \frac{7}{9}$
 Note: $E(X^2) = m''(0) = \frac{2}{(3-0)^3} + \frac{2}{3(1-0)^3} = \frac{10}{27}$

MGF: $M_X(t) = \begin{cases} \frac{\lambda}{\lambda-t} & \text{if } t < \lambda \\ \text{undef} & \text{else} \end{cases}$ Note: $M_{X+Y}(t) = M_X(t) M_Y(t)$
 density: $f_X(x) = \begin{cases} \frac{\lambda}{2} e^{-\lambda x} & \text{if } x < 0 \\ \frac{\lambda}{2} e^{-\lambda x} & \text{if } x \geq 0 \end{cases}$
 a) $M_X(t) = E(e^{tX}) = \int_{-\infty}^{\infty} e^{tx} f_X(x) dx$
 $= \int_{-\infty}^0 e^{tx} \frac{\lambda}{2} e^{-\lambda x} dx + \int_0^{\infty} e^{tx} \frac{\lambda}{2} e^{-\lambda x} dx$
 $= \frac{\lambda}{2} \left(\int_{-\infty}^0 e^{(t-\lambda)x} dx + \int_0^{\infty} e^{(t-\lambda)x} dx \right)$
 $= \frac{\lambda}{2} \left(\left[\frac{e^{(t-\lambda)x}}{t-\lambda} \right]_{-\infty}^0 + \left[\frac{e^{(t-\lambda)x}}{t-\lambda} \right]_0^{\infty} \right)$
 $= \frac{\lambda}{2} \left(\frac{1}{t-\lambda} - \frac{1}{t-\lambda} \right) = \frac{\lambda}{2} \left(\frac{1}{(t-\lambda)(3-t)} \right)$

$E(X) = \int_0^{\infty} 1 - \frac{y^4}{16} dy = 1 - \frac{y^5}{80} \Big|_0^1 = 1 - \frac{1}{80} = \frac{79}{80}$
 $E(X) = \int_0^{\infty} 1 - F(x) dx$

$M_{U \sim V}(u) = M_U(u) \cdot M_V(-u)$ $U \sim \text{Exp}(\lambda)$
 $= \left(\frac{\lambda}{\lambda-u} \right) \left(\frac{\lambda}{\lambda-(-u)} \right) = \frac{\lambda^2}{(\lambda-u)(\lambda+u)} \Rightarrow \text{Jame}$

$Z = Y - X$, $P(Z \leq 0.5) = \frac{1 - e^{-0.5\lambda}}{1 - e^{-\lambda}}$, $X \sim \text{Exp}(\lambda)$
 $\sum_{n=0}^{\infty} ar^n = a + ar + ar^2 + \dots = \frac{a}{1-r}$ rounds down at 0.5
 a) $P(Z \leq 0.5) = \sum_{k=0}^{\infty} P(k \leq X \leq k+0.5)$
 $= \sum_{k=0}^{\infty} (P(Z > k) - P(Z > k+0.5))$
 $= \sum_{k=0}^{\infty} (e^{-k\lambda} - e^{-(k+0.5)\lambda}) = \sum_{k=0}^{\infty} e^{-k\lambda} - \sum_{k=0}^{\infty} e^{-(k+0.5)\lambda} = \frac{1}{1-e^{-\lambda}} - \frac{e^{-0.5\lambda}}{1-e^{-\lambda}}$
 $\rightarrow$ PDF $\rightarrow$ int $\rightarrow$ (DF) "diff" gives $f(x) \leftarrow$ PDF

Prac Quiz 3

1) Find $\text{var}(X)$
 a) X w/ values in $[0,1]$ & $f_X(x) = Cx^2(1-x)^3$ for $0 < x < 1$
 $X \sim \text{Beta}(r=3, s=4)$
 $\text{var}(X) = \frac{rs}{(r+s)^2(r+s+1)} = \frac{3 \cdot 4}{(3+4)^2(3+4+1)} = \frac{12}{49 \cdot 8} = \frac{3}{98}$

b) X w/ values in $[0, \infty)$ & density $C_2 x^4 e^{-x}$ for $0 \leq x < \infty$
 $X \sim \text{Gamma}(r=5, \lambda=1)$
 $\text{Var}(X) = \frac{r}{\lambda^2} = \frac{5}{1} = 5$

2) $X \sim \text{unif}[-\frac{\pi}{2}, \pi]$, $Y = \sin(X)$
 a) Find $f_Y(-\frac{1}{2})$
 i) $f_X(x) = \frac{1}{b-a} = \frac{1}{\frac{3\pi}{2}}$ for $x \in [-\frac{\pi}{2}, \pi]$
 ii) $y = \sin(x)$ (not 1 to 1)
 $y \in [0, 1]$ gives 2 sol in $[-\frac{\pi}{2}, \pi]$:
 $\rightarrow x_1 = \arcsin(y)$
 $\rightarrow x_2 = \pi - \arcsin(y)$
 $y \in [0, -1]$ gives 1 sol:
 $\rightarrow x = \arcsin(y)$

3) $Y = \sin(X)$ • $x_2 = \pi - \arcsin(y)$
 • $X = \arcsin(y)$ $\left| \frac{dx}{dy} \right| = \frac{1}{\sqrt{1-y^2}}$
 • $f_Y(y) = \sum_{X: \sin(X)=y} \frac{f_X(X)}{\left| \frac{dx}{dy} \right|} = \sum_{X: \sin(X)=y} f_X(X) \left| \frac{dx}{dy} \right|$
 $f_Y(\frac{1}{2}) = \frac{1}{\frac{3\pi}{2}} \cdot \sqrt{1-\frac{1}{4}} = \frac{2}{3\pi} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{3\pi}$
 $f_Y(\frac{1}{2}) \rightarrow$ plug in (hds 2 that you add)

4) All together:
 $f_Y(y) = \begin{cases} \frac{2}{3\pi} \cdot \frac{1}{\sqrt{1-y^2}}, & y \in (-1, 1) \\ 0, & y \in \{-1, 1\} \end{cases}$

Note: $r(n) = (n-1)!$

Beta(r, s) # of gaps after $n = r+s-1$
 # of gaps before $n = r+s-1$
 total # of gaps $n = r+s-1$
 total # of words



5) $Y = \sin(X)$ • $x_2 = \pi - \arcsin(y)$
 • $X = \arcsin(y)$ $\left| \frac{dx}{dy} \right| = \frac{1}{\sqrt{1-y^2}}$
 • $f_Y(y) = \sum_{X: \sin(X)=y} \frac{f_X(X)}{\left| \frac{dx}{dy} \right|} = \sum_{X: \sin(X)=y} f_X(X) \left| \frac{dx}{dy} \right|$
 $f_Y(\frac{1}{2}) = \frac{1}{\frac{3\pi}{2}} \cdot \sqrt{1-\frac{1}{4}} = \frac{2}{3\pi} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{3\pi}$
 $f_Y(\frac{1}{2}) \rightarrow$ plug in (hds 2 that you add)

6) All together:
 $f_Y(y) = \begin{cases} \frac{2}{3\pi} \cdot \frac{1}{\sqrt{1-y^2}}, & y \in (-1, 1) \\ 0, & y \in \{-1, 1\} \end{cases}$